FGFR3 fusion-hotspot benchmark report

Study: msk_impact_50k_2026

Abstract

This report analyzes FGFR3 gene fusions from the msk_impact_50k_2026 study, covering 152 fusion events. 74/152 fusions (48.7%) were in-frame. 142/152 fusions (93.4%) retained the PF07714 domain. Domain retention was tested with Fisher's exact test (p=0.0194824, statistically significant at alpha=0.05). No literature benchmark is configured for FGFR3.

Results summary

Fusion partner frequency

Fusion-partner frequency was tabulated across 152 analyzed fusion events, identifying 15 distinct fusion partner genes. The most frequent partner was TACC3, observed in 131 events.

Domain retention

PF07714 domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 73/74 (98.6%) of in-frame fusions retained the domain, p=0.0194824 (statistically significant at alpha=0.05). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.00990099.

Domain disruption

Domain-disruption analysis was skipped: FGFR3 has no disruption_required_domains configured; domain-disruption analysis was skipped.

Cutpoint detection

Cutpoint detection scanned 151 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 677 aa (permutation-corrected p=0.00990099, statistically significant at alpha=0.05). This is -71 aa from the nearest configured domain boundary at 748 aa.

Corroborating confidence statistics

Corroborating confidence statistics were not computed for this run: FGFR3 has no group_field configured; confidence-stats analysis was skipped.

Composite evidence score

Composite evidence score reported: domain_disruption sub-score excluded: the supplied result had no applicable p-value (e.g. disruption_required_domains not configured for this gene).

Exon retention

Exon retention reported: No target exon was supplied in params or GeneConfig.

Joint partner

Joint partner reported: FGFR3 has no gene_pair configured; joint-partner analysis was skipped.

Tables

composite_score tables

composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
TACC3	131	0.8618421052631579	0.20043213737826426	None	0.8507823563952885	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.6232543134040097	1
TNIP2	3	0.019736842105263157	0.20043213737826426	None	0.8763197586726998	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.26782435825292855	2
CCDC149	2	0.013157894736842105	0.20043213737826426	None	0.8778280542986425	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2653280155863073	3
JAKMIP1	2	0.013157894736842105	0.20043213737826426	None	0.8778280542986425	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2653280155863073	4
LPCAT2	2	0.013157894736842105	0.20043213737826426	None	0.8770739064856712	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.26516641248352774	5
PTBP1	1	0.006578947368421052	0.20043213737826426	None	0.8778280542986425	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2625084667141268	6
CTBP1	1	0.006578947368421052	0.20043213737826426	None	0.8763197586726998	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.26218526050856766	7
MAEA	1	0.006578947368421052	0.20043213737826426	None	0.8506787330316742	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2566907550140622	8
ROBO1	1	0.006578947368421052	0.20043213737826426	None	0.8159879336349924	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.24925701228620178	9
NSD2	3	0.019736842105263157	0.20043213737826426	None	0.6561085972850679	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.22063625224129316	10
RBM20	1	0.006578947368421052	0.20043213737826426	None	0.2413273001508296	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.1261154479681669	11
BABAM2	1	0.006578947368421052	0.20043213737826426	None	0.20965309200603321	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.11932811765142481	12
SLIT2	1	0.006578947368421052	0.20043213737826426	None	None	None	['domain_retention', 'recurrence']	0.0946940337365316	13
GRK4	1	0.006578947368421052	0.20043213737826426	None	0.0015082956259426794	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.07472566128426256	14

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
DGKQ	1	0.006578947368421052	0.20043213737826426	None	0.0	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.07440245507870341	15

cutpoint_detection tables

cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
14	1	141	0	9	None	1.0	-0.0
15	2	140	0	9	None	1.0	-0.0
42	2	140	1	8	0.11428571428571428	0.1694208631494733	0.7710331099865381
127	3	139	1	8	0.17266187050359713	0.21992905390389725	0.6577173940286984
153	4	138	1	8	0.2318840579710145	0.26768849958325047	0.5723702865449334
174	4	138	2	7	0.10144927536231885	0.04197605082466328	1.3769984229650098
188	4	138	3	6	0.057971014492753624	0.004616748961172982	2.33566373986112
311	4	138	4	5	0.036231884057971016	0.00037923555536042514	3.421090951996021
376	4	138	5	4	0.02318840579710145	2.3640024802728035e-05	4.626352072135864
648	4	138	6	3	0.014492753623188406	1.1022799082261042e-06	5.957708108603845
653	4	138	7	2	0.008281573498964804	3.6665492410410534e-08	7.4357424781390336
677	4	138	8	1	0.0036231884057971015	7.825580002902196e-10	9.106483464295888
753	5	137	8	1	0.004562043795620438	2.021680915172072e-09	8.694287388663584
758	52	90	8	1	0.07222222222222222	0.002807318578004277	2.5517083002909002
759	88	54	8	1	0.2037037037037037	0.15623087834924132	0.806233125571681
760	89	53	8	1	0.2099056603773585	0.15840511374693833	0.8002308023157874
762	91	51	8	1	0.22303921568627452	0.16500164368454323	0.7825117294856598
763	93	49	8	1	0.2372448979591837	0.27269821801700755	0.5643177000044979
764	95	47	8	1	0.2526595744680851	0.2731204045238703	0.5636458529869772
766	96	46	8	1	0.2608695652173913	0.27473748927465147	0.5610820749288234
767	97	45	8	1	0.26944444444444443	0.27730934214956404	0.5570354994244366
768	100	42	8	1	0.2976190476190476	0.4468103647595553	0.34987676101374454
769	101	41	8	1	0.3079268292682927	0.4455794616522395	0.3510748355557522
770	105	37	8	1	0.3547297297297297	0.45054685984922543	0.34626003284121815
771	108	34	8	1	0.39705882352941174	0.6854647233823904	0.16401489074257258
772	111	31	8	1	0.4475806451612903	0.6849966189342125	0.16431157213069608
773	113	29	8	1	0.4870689655172414	0.688608956187972	0.16202733307235542

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
776	115	27	8	1	0.5324074074074074	1.0	-0.0
777	119	23	8	1	0.6467391304347826	1.0	-0.0
778	120	22	8	1	0.6818181818181818	1.0	-0.0
779	121	21	8	1	0.7202380952380952	1.0	-0.0
780	122	20	8	1	0.7625	1.0	-0.0
781	125	17	8	1	0.9191176470588235	1.0	-0.0
782	127	15	8	1	1.0583333333333333	1.0	-0.0
785	128	14	8	1	1.1428571428571428	1.0	-0.0
786	130	12	8	1	1.3541666666666667	0.5655253555991285	0.24754791850170255
787	131	11	9	0	0.0	1.0	-0.0
788	133	9	9	0	0.0	1.0	-0.0
789	134	8	9	0	0.0	1.0	-0.0
793	138	4	9	0	0.0	1.0	-0.0
797	139	3	9	0	0.0	1.0	-0.0
799	140	2	9	0	0.0	1.0	-0.0

domain_retention tables

frame_domain_contingency_table

73	69
1	8

domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
kinase	Protein tyrosine and serine/threonine kinase	151	142	3	6	0.22663457681508226

frequency tables

Partner_gene_counts

Partner_gene	Event_count
BABAM2	1
CCDC149	2
CTBP1	1

Partner_gene	Event_count
DGKQ	1
GRK4	1
JAKMIP1	2
LPCAT2	2
MAEA	1
NSD2	3
PTBP1	1
RBM20	1
ROBO1	1
SLIT2	1
TACC3	131
TNIP2	3

Per-event results (results.tsv)

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0000025-T03-IM6-1	P-0000025-T03-IM6		DGKQ-FGFR3	DGKQ	unknown	three_prime	1795701	1	14	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase			DGKQ (NM_001347) - FGFR3 (NM_000142) rearrangement: c.352-750:DGKQ_c.40:FGFR3inv | Protein Fusion: mid-exon {DGKQ:FGFR3}	NA
EVT-P-0020773-T01-IM6-2	P-0020773-T01-IM6		FGFR3-BABAM2	BABAM2	unknown	three_prime	1803105	4	153	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase			BRE (NM_004899) - FGFR3 (NM_000142) rearrangement: t(2;4)(p23.2;p16.3)(chr2:g.28322454::chr4:g.1803105) | Protein Fusion: mid-exon {BRE:FGFR3}	NA
EVT-P-0054353-T02-IM7-4	P-0054353-T02-IM7		FGFR3-CCDC149	CCDC149	in-frame	five_prime	1808701	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - CCDC149 (NM_001130726) fusion: c.2274+40:FGFR3_c.490-99:CCDC149inv | Protein Fusion: in frame {FGFR3:CCDC149}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0054353-T01-IM6-5	P-0054353-T01-IM6		FGFR3-CCDC149	CCDC149	in-frame	five_prime	1808701	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - CCDC149 (NM_001130726) fusion: c.2274+40:FGFR3_c.490-99:CCDC149inv | Protein Fusion: in frame {FGFR3:CCDC149}	NA
EVT-P-0057559-T02-IM7-6	P-0057559-T02-IM7		FGFR3-CTBP1	CTBP1	in-frame	five_prime	1808812	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - CTBP1 (NM_001328) fusion: c.2275-31:FGFR3_c.196-1497:CTBP1inv | Protein Fusion: in frame {FGFR3:CTBP1}	NA
EVT-P-0019022-T02-IM6-35	P-0019022-T02-IM6		FGFR3-GRK4	GRK4	unknown	three_prime	1795705	1	15	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase			GRK4 (NM_182982) - FGFR3 (NM_000142) rearrangement: c.339+3191:GRK4_c.44:FGFR3dup | Protein Fusion: mid-exon {GRK4:FGFR3}	NA
EVT-P-0002610-T01-IM3-36	P-0002610-T01-IM3		FGFR3-JAKMIP1	JAKMIP1	in-frame	five_prime	1808734	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			NA | Protein fusion: in frame (FGFR3-JAKMIP1)	NA
EVT-P-0034419-T01-IM6-37	P-0034419-T01-IM6		FGFR3-JAKMIP1	JAKMIP1	in-frame	five_prime	1808749	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - JAKMIP1 (NM_001099433) fusion (FGFR3 exons 1-17 fused to JAKM1P1 exons 4-21): c.2274+88:FGFR3_c.624+9161:JAKMIP1inv | Protein Fusion: in frame {FGFR3:JAKMIP1}	NA
EVT-P-0068474-T01-IM7-39	P-0068474-T01-IM7		FGFR3-MAEA	MAEA	unknown	five_prime	1808895	17	776	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - MAEA (NM_001017405) fusion: c.2327:FGFR3_c.457-2460:MAEAdup | Protein Fusion: mid-exon {FGFR3:MAEA}	NA
EVT-P-0010465-T01-IM5-42	P-0010465-T01-IM5		FGFR3-NSD2	NSD2	unknown	five_prime	1807917	13	653	True	disrupted	0.65703971119133 58	True			Protein tyrosine and serine/threonine kinase	FGFR3 (NM_000142) - WHSC1 (NM_001042424) rearrangement:c.1959+17:FGFR3_c.-29-12740:WHSC1del | Transcript fusion (FGFR3-WHSC1)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0013954-T01-IM5-43	P-0013954-T01-IM5		FGFR3-NSD2	NSD2	unknown	three_prime	1808929	17	787	False	lost	0.0	False		Protein tyrosine and serine/threonine kinase		WHSC1 (NM_001042424) - FGFR3 (NM_000142) rearrangement: c.3255+4: WHSC1_c.2361:FGFR3dup | Protein fusion: mid-exon (WHSC1-FGFR3)	NA
EVT-P-0024991-T01-IM6-44	P-0024991-T01-IM6		FGFR3-ROBO1	ROBO1	unknown	five_prime	1808963	17	799	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - ROBO1 (NM_002941) fusion: t(3;4)(p12.3;p16.3)(chr3:g.78737098::chr4:g.1808963) | Protein Fusion: mid-exon {FGFR3:ROBO1}	NA
EVT-P-0022387-T02-IM6-48	P-0022387-T02-IM6		FGFR3-TACC3	TACC3	in-frame	five_prime	1808707	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2274+46:FGFR3_c.1942-224:TACC3dup | Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-0044466-T02-IM6-49	P-0044466-T02-IM6		FGFR3-TACC3	TACC3	in-frame	five_prime	1808773	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2275-70:FGFR3_c.1645-159:TACC3dup | Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-0055884-T01-IM6-50	P-0055884-T01-IM6		FGFR3-TACC3	TACC3	unknown	five_prime	1808886	17	773	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2318:FGFR3_c.1942-928:TACC3dup | Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-0000941-T02-IM6-51	P-0000941-T02-IM6		FGFR3-TACC3	TACC3	unknown	five_prime	1808894	17	776	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2326:FGFR3_c.1942-988:TACC3dup | Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-0056846-T01-IM6-52	P-0056846-T01-IM6		FGFR3-TACC3	TACC3	in-frame	five_prime	1808698	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2274+37:FGFR3_c.1942-426:TACC3dup | Protein Fusion: in frame {FGFR3:TACC3}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0053694-T03-IM6-53	P-0053694-T03-IM6		FGFR3-TACC3	TACC3	unknown	five_prime	1808911	17	781	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2343:FGFR3_c.1942-151:TACC3dup | Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-0053694-T02-IM6-54	P-0053694-T02-IM6		FGFR3-TACC3	TACC3	unknown	five_prime	1808911	17	781	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2343:FGFR3_c.1942-151:TACC3dup | Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-0016989-T02-IM6-55	P-0016989-T02-IM6		FGFR3-TACC3	TACC3	in-frame	five_prime	1808718	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2274+57:FGFR3_c.1645-139:TACC3dup | Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-0053694-T01-IM6-57	P-0053694-T01-IM6		FGFR3-TACC3	TACC3	unknown	five_prime	1808911	17	781	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2343:FGFR3_c.1942-151:TACC3dup | Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-0052352-T02-IM6-59	P-0052352-T02-IM6		FGFR3-TACC3	TACC3	in-frame	five_prime	1808764	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2275-79:FGFR3_c.1837-58:TACC3dup | Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-0052324-T01-IM6-60	P-0052324-T01-IM6		FGFR3-TACC3	TACC3	in-frame	five_prime	1808803	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2275-40:FGFR3_c.1942-486:TACC3dup | Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-0059195-T01-IM7-61	P-0059195-T01-IM7		FGFR3-TACC3	TACC3	in-frame	five_prime	1808841	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2275-2:FGFR3_c.1645-51:TACC3dup | Protein Fusion: in frame {FGFR3:TACC3}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0060831-T01-IM7-62	P-0060831-T01-IM7		FGFR3-TACC3	TACC3	in-frame	five_prime	1808827	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2275-16:FGFR3_c.1941+414:TACC3dup | Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-0060879-T01-IM7-63	P-0060879-T01-IM7		FGFR3-TACC3	TACC3	unknown	five_prime	1808945	17	793	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2377:FGFR3_c.1942-241:TACC3dup | Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-0011503-T01-IM5-64	P-0011503-T01-IM5		FGFR3-TACC3	TACC3	unknown	five_prime	1808967	17	800	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-18 with TACC3 exons 10-16): c.2399:FGFR3_c.1836+75:TACC3dup | Protein fusion: mid-exon (FGFR3-TACC3)	NA
EVT-P-0051755-T01-IM6-65	P-0051755-T01-IM6		FGFR3-TACC3	TACC3	in-frame	five_prime	1808806	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2275-37:FGFR3_c.1942-333:TACC3dup | Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-0010534-T01-IM5-66	P-0010534-T01-IM5		FGFR3-TACC3	TACC3	unknown	five_prime	1808859	17	764	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-18 fused to TACC3 exons 10-16): c.2363:FGFR3_c.1784:TACC3dup | Protein fusion: mid-exon (FGFR3-TACC3)	NA
EVT-P-0010096-T03-IM6-67	P-0010096-T03-IM6		FGFR3-TACC3	TACC3	unknown	five_prime	1808925	17	786	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2357:FGFR3_c.1941+378:TACC3dup | Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-0060879-T02-IM7-68	P-0060879-T02-IM7		FGFR3-TACC3	TACC3	unknown	five_prime	1808945	17	793	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2377:FGFR3_c.1942-241:TACC3dup | Protein Fusion: mid-exon {FGFR3:TACC3}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0062326-T01-IM7-69	P-0062326-T01-IM7		FGFR3-TACC3	TACC3	unknown	five_prime	1808913	17	782	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2345:FGFR3_c.1386-12:TACC3dup | Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-0051602-T01-IM6-70	P-0051602-T01-IM6		FGFR3-TACC3	TACC3	unknown	five_prime	1808884	17	772	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2316:FGFR3_c.1942-997:TACC3dup | Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-0051575-T01-IM6-71	P-0051575-T01-IM6		FGFR3-TACC3	TACC3	in-frame	five_prime	1808724	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2274+63:FGFR3_c.1461+64:TACC3dup | Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-0051267-T01-IM6-72	P-0051267-T01-IM6		FGFR3-TACC3	TACC3	in-frame	five_prime	1808672	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2274+11:FGFR3_c.1645-143:TACC3dup | Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-0051173-T02-IM6-73	P-0051173-T02-IM6		FGFR3-TACC3	TACC3	in-frame	five_prime	1808716	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2274+55:FGFR3_c.1941+61:TACC3dup | Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-0063306-T01-IM7-74	P-0063306-T01-IM7		FGFR3-TACC3	TACC3	in-frame	five_prime	1808764	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2275-79:FGFR3_c.1461+18:TACC3dup | Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-0063929-T01-IM7-75	P-0063929-T01-IM7		FGFR3-TACC3	TACC3	in-frame	five_prime	1808782	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2275-61:FGFR3_c.1941+145:TACC3dup | Protein Fusion: in frame {FGFR3:TACC3}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0051724-T01-IM6-76	P-0051724-T01-IM6		FGFR3-TACC3	TACC3	in-frame	five_prime	1808760	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2275-83:FGFR3_c.1942-112:TACC3dup | Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-0050757-T02-IM7-77	P-0050757-T02-IM7		FGFR3-TACC3	TACC3	unknown	five_prime	1808660	16	758	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2273:FGFR3_c.1572:TACC3dup | Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-0006879-T01-IM5-78	P-0006879-T01-IM5		FGFR3-TACC3	TACC3	unknown	five_prime	1808967	17	800	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) Fusion (FGFR3 exons 1 to 18 fused with TACC3 exons 11-16) : c.2399:FGFR3_c.1941+196:TACC3dup | Protein fusion: mid-exon (FGFR3-TACC3)	NA
EVT-P-0025859-T01-IM6-79	P-0025859-T01-IM6		FGFR3-TACC3	TACC3	unknown	five_prime	1800997	2	42	False	lost	0.0	False		Protein tyrosine and serine/threonine kinase		FGFR3 (NM_000142) - TACC3 (NM_006342) rearrangement: c.126:FGFR3_c.1941+33:TACC3dup | Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-0050136-T01-IM6-80	P-0050136-T01-IM6		FGFR3-TACC3	TACC3	unknown	five_prime	1808922	17	785	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2354:FGFR3_c.2040:TACC3dup | Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-0027054-T03-IM6-81	P-0027054-T03-IM6		FGFR3-TACC3	TACC3	unknown	five_prime	1808898	17	777	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2330:FGFR3_c.1942-902:TACC3dup | Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-0027054-T05-IM6-82	P-0027054-T05-IM6		FGFR3-TACC3	TACC3	unknown	five_prime	1808898	17	777	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2330:FGFR3_c.1942-902:TACC3dup | Protein Fusion: mid-exon {FGFR3:TACC3}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0027054-T07-IM7-83	P-0027054-T07-IM7		FGFR3-TACC3	TACC3	unknown	five_prime	1808898	17	777	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2330:FGFR3_c.1942-902:TACC3dup | Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-0049741-T02-IM6-84	P-0049741-T02-IM6		FGFR3-TACC3	TACC3	in-frame	five_prime	1808830	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2275-13:FGFR3_c.1942-479:TACC3dup | Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-0049502-T01-IM6-85	P-0049502-T01-IM6		FGFR3-TACC3	TACC3	in-frame	five_prime	1808726	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2274+65:FGFR3_c.1941+538:TACC3dup | Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-0048985-T01-IM6-86	P-0048985-T01-IM6		FGFR3-TACC3	TACC3	unknown	five_prime	1808827	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2275-16:FGFR3_c.1788:TACC3dup | Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-0048947-T01-IM6-87	P-0048947-T01-IM6		FGFR3-TACC3	TACC3	unknown	five_prime	1808898	17	777	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2330:FGFR3_c.1941+104:TACC3dup | Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-0006660-T02-IM6-88	P-0006660-T02-IM6		FGFR3-TACC3	TACC3	out-of-frame	five_prime	1808678	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2274+17:FGFR3_c.1591+1556:TACC3dup | Protein Fusion: out of frame {FGFR3:TACC3}	NA
EVT-P-0047604-T01-IM6-89	P-0047604-T01-IM6		FGFR3-TACC3	TACC3	in-frame	five_prime	1808692	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2274+31:FGFR3_c.1941+484:TACC3dup | Protein Fusion: in frame {FGFR3:TACC3}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0045507-T01-IM6-90	P-0045507-T01-IM6		FGFR3-TACC3	TACC3	in-frame	five_prime	1808733	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2274+72:FGFR3_c.1942-413:TACC3dup | Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-0045382-T01-IM6-91	P-0045382-T01-IM6		FGFR3-TACC3	TACC3	unknown	five_prime	1808903	17	779	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2335:FGFR3_c.1942-101:TACC3dup | Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-0047573-T01-IM6-92	P-0047573-T01-IM6		FGFR3-TACC3	TACC3	unknown	five_prime	1808844	17	759	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2276:FGFR3_c.1942-892:TACC3dup | Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-0064650-T01-IM7-93	P-0064650-T01-IM7		FGFR3-TACC3	TACC3	in-frame	five_prime	1808781	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2275-62:FGFR3_c.1837-81:TACC3dup | Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-0065466-T01-IM7-94	P-0065466-T01-IM7		FGFR3-TACC3	TACC3	in-frame	five_prime	1808795	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2275-48:FGFR3_c.1942-685:TACC3dup | Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-0005361-T01-IM5-96	P-0005361-T01-IM5		FGFR3-TACC3	TACC3	unknown	five_prime	1803212	4	188	False	lost	0.0	False		Protein tyrosine and serine/threonine kinase		FGFR3 (NM_000142) -TACC3 (NM_006342) Fusion (FGFR3 exon 5 fused to TACC3 exon 14) : c.564:FGFR3_c.2223+249:TACC3dup | Protein fusion: mid-exon (FGFR3-TACC3)	NA
EVT-P-0005128-T01-IM5-97	P-0005128-T01-IM5		FGFR3-TACC3	TACC3	in-frame	five_prime	1808814	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3(NM_000142) -TACC3 (NM_006342) Fusion (FGFR3 exon 17 fused to TACC3 exon 11) : c.2275-29:FGFR3_c.1942-314:TACC3dup | Protein fusion: in frame (FGFR3-TACC3)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0032266-T01-IM6-98	P-0032266-T01-IM6		FGFR3-TACC3	TACC3	in-frame	five_prime	1808745	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2274+84:FGFR3_c.1645-165:TACC3dup | Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-0016238-T03-IM7-99	P-0016238-T03-IM7		FGFR3-TACC3	TACC3	in-frame	five_prime	1808706	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2274+45:FGFR3_c.1942-654:TACC3dup | Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-0047604-T02-IM6-100	P-0047604-T02-IM6		FGFR3-TACC3	TACC3	in-frame	five_prime	1808692	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2274+31:FGFR3_c.1941+484:TACC3dup | Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-0066401-T01-IM7-101	P-0066401-T01-IM7		FGFR3-TACC3	TACC3	unknown	five_prime	1808930	17	788	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2362:FGFR3_c.1942-935:TACC3dup | Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-0004694-T01-IM5-102	P-0004694-T01-IM5		FGFR3-TACC3	TACC3	unknown	five_prime	1808881	17	771	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3(NM_000142) -TACC3(NM_006342) Fusion (FGFR3 exon 17 fused with TACC3 exon 9) : c.2313:FGFR3_c.1837-88:TACC3dup | Protein fusion: mid-exon (FGFR3-TACC3)	NA
EVT-P-0003853-T01-IM5-103	P-0003853-T01-IM5		FGFR3-TACC3	TACC3	in-frame	five_prime	1808680	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1 to 17 with TACC3 exons 11 to 16) | Protein fusion: in frame (FGFR3-TACC3)	NA
EVT-P-0003827-T01-IM5-104	P-0003827-T01-IM5		FGFR3-TACC3	TACC3	in-frame	five_prime	1808714	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-17 fused to TACC3 exons 11-16): c.1941+232:TACC3_c.2274+53:FGFR3dup | Protein fusion: in frame (FGFR3-TACC3)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0046580-T03-IM6-105	P-0046580-T03-IM6		FGFR3-TACC3	TACC3	unknown	five_prime	1808716	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2274+55:FGFR3_c.1943:TACC3dup | Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-0046580-T02-IM6-106	P-0046580-T02-IM6		FGFR3-TACC3	TACC3	unknown	five_prime	1808716	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2274+55:FGFR3_c.1943:TACC3dup | Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-0045507-T04-IM7-107	P-0045507-T04-IM7		FGFR3-TACC3	TACC3	in-frame	five_prime	1808733	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2274+72:FGFR3_c.1942-413:TACC3dup | Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-0037126-T01-IM6-108	P-0037126-T01-IM6		FGFR3-TACC3	TACC3	unknown	five_prime	1808878	17	770	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2310:FGFR3_c.1837-16:TACC3dup | Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-0038253-T01-IM6-109	P-0038253-T01-IM6		FGFR3-TACC3	TACC3	unknown	five_prime	1808852	17	762	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) -TACC3(NM_006342) Fusion (FGFR3 exon18 fused with TACC3 exon8) : c.2284:FGFR3_c.1644+106:TACC3dup | Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-0038318-T01-IM6-110	P-0038318-T01-IM6		FGFR3-TACC3	TACC3	in-frame	five_prime	1808756	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2275-87:FGFR3_c.1462-57:TACC3dup | Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-0038414-T01-IM6-111	P-0038414-T01-IM6		FGFR3-TACC3	TACC3	unknown	five_prime	1808869	17	767	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2301:FGFR3_c.1941+974:TACC3dup | Protein Fusion: mid-exon {FGFR3:TACC3}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0039109-T01-IM6-112	P-0039109-T01-IM6		FGFR3-TACC3	TACC3	in-frame	five_prime	1808760	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2275-83:FGFR3_c.162+65:TACC3dup | Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-0039109-T02-IM6-113	P-0039109-T02-IM6		FGFR3-TACC3	TACC3	in-frame	five_prime	1808760	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2275-83:FGFR3_c.162+65:TACC3dup | Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-0039116-T01-IM6-114	P-0039116-T01-IM6		FGFR3-TACC3	TACC3	unknown	five_prime	1808698	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2274+37:FGFR3_c.944:TACC3dup | Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-0039138-T02-IM6-115	P-0039138-T02-IM6		FGFR3-TACC3	TACC3	unknown	five_prime	1808645	16	753	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2258:FGFR3_c.1941+335:TACC3dup | Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-0039166-T01-IM6-116	P-0039166-T01-IM6		FGFR3-TACC3	TACC3	unknown	five_prime	1808883	17	772	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2315:FGFR3_c.1942-61:TACC3dup | Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-0067303-T01-IM7-117	P-0067303-T01-IM7		FGFR3-TACC3	TACC3	in-frame	five_prime	1808812	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2275-31:FGFR3_c.1941+863:TACC3dup | Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-0039655-T01-IM6-118	P-0039655-T01-IM6		FGFR3-TACC3	TACC3	unknown	five_prime	1808887	17	773	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2319:FGFR3_c.1836+86:TACC3dup | Protein Fusion: mid-exon {FGFR3:TACC3}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0003279-T02-IM6-119	P-0003279-T02-IM6		FGFR3-TACC3	TACC3	in-frame	five_prime	1808708	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) rearrangement: c.2274+47:FGFR3_c.1941+850:TACC3dup | Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-0040543-T01-IM6-120	P-0040543-T01-IM6		FGFR3-TACC3	TACC3	in-frame	five_prime	1808696	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2274+35:FGFR3_c.1836+39:TACC3dup | Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-0067315-T01-IM7-121	P-0067315-T01-IM7		FGFR3-TACC3	TACC3	unknown	five_prime	1808877	17	770	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2309:FGFR3_c.1941+162:TACC3dup | Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-0067602-T01-IM7-122	P-0067602-T01-IM7		FGFR3-TACC3	TACC3	in-frame	five_prime	1808775	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2275-68:FGFR3_c.1836+111:TACC3dup | Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-0047905-T01-IM6-123	P-0047905-T01-IM6		FGFR3-TACC3	TACC3	in-frame	five_prime	1808802	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2275-41:FGFR3_c.1837-96:TACC3dup | Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-0042157-T06-IM7-124	P-0042157-T06-IM7		FGFR3-TACC3	TACC3	unknown	five_prime	1808855	17	763	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2287:FGFR3_c.1942-923:TACC3dup | Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-0042393-T01-IM6-125	P-0042393-T01-IM6		FGFR3-TACC3	TACC3	in-frame	five_prime	1808709	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2274+48:FGFR3_c.1644+132:TACC3dup | Protein Fusion: in frame {FGFR3:TACC3}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0042515-T01-IM6-126	P-0042515-T01-IM6		FGFR3-TACC3	TACC3	in-frame	five_prime	1808841	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2275-2:FGFR3_c.1942-418:TACC3dup | Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-0043076-T01-IM6-127	P-0043076-T01-IM6		FGFR3-TACC3	TACC3	unknown	five_prime	1808926	17	786	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2358:FGFR3_c.2019-34:TACC3dup | Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-0044057-T01-IM6-128	P-0044057-T01-IM6		FGFR3-TACC3	TACC3	in-frame	five_prime	1808714	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2274+53:FGFR3_c.1645-189:TACC3dup | Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-0016624-T01-IM6-130	P-0016624-T01-IM6		FGFR3-TACC3	TACC3	unknown	five_prime	1808733	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exon 17 fused to TACC3 exon 5): c.2274+72:FGFR3_c.1387:TACC3dup | Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-0045507-T03-IM7-131	P-0045507-T03-IM7		FGFR3-TACC3	TACC3	in-frame	five_prime	1808733	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2274+72:FGFR3_c.1942-413:TACC3dup | Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-0054391-T01-IM6-132	P-0054391-T01-IM6		FGFR3-TACC3	TACC3	in-frame	five_prime	1808789	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2275-54:FGFR3_c.1941+447:TACC3dup | Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-0006429-T01-IM5-133	P-0006429-T01-IM5		FGFR3-TNIP2	TNIP2	unknown	five_prime	1808801	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) -TNIP2 (NM_024309) rearrangement: c.2275-42:FGFR3_c.147:TNIP2inv | Protein fusion: mid-exon (FGFR3-TNIP2)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0006429-T02-IM6-134	P-0006429-T02-IM6		FGFR3-TNIP2	TNIP2	unknown	five_prime	1808753	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TNIP2 (NM_024309) fusion: c.2275-90:FGFR3_c.195:TNIP2inv | Protein Fusion: mid-exon {FGFR3:TNIP2}	NA
EVT-P-0050971-T01-IM6-135	P-0050971-T01-IM6		FGFR3-TNIP2	TNIP2	unknown	five_prime	1808827	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TNIP2 (NM_024309) fusion: c.2275-16:FGFR3_c.252:TNIP2inv | Protein Fusion: mid-exon {FGFR3:TNIP2}	NA
EVT-P-0012257-T01-IM5-136	P-0012257-T01-IM5		LPCAT2-FGFR3	LPCAT2	in-frame	five_prime	1808759	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - LPCAT2 (NM_017839) rearrangement: t(4;16)(p16.3;q12.2)(chr4:g.1808759::chr16:g.55556795) | Protein fusion: in frame (FGFR3-LPCAT2)	NA
EVT-P-0012257-T03-IM6-137	P-0012257-T03-IM6		LPCAT2-FGFR3	LPCAT2	in-frame	five_prime	1808708	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - LPCAT2 (NM_017839) rearrangement: t(4;16)(p16.3;q12.2)(chr4:g.1808708::chr16:g.55556796) | Protein Fusion: in frame {FGFR3:LPCAT2}	NA
EVT-P-0024146-T01-IM6-139	P-0024146-T01-IM6		NSD2-FGFR3	NSD2	in-frame	three_prime	1801414	3	127	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			WHSC1 (NM_001042424) - FGFR3 (NM_000142) rearrangement: c.761-178:WHSC1_c.380-60:FGFR3dup | Protein Fusion: in frame {WHSC1:FGFR3}	NA
EVT-P-0017789-T01-IM6-141	P-0017789-T01-IM6		PTBP1-FGFR3	PTBP1	in-frame	five_prime	1808676	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - PTBP1 (NM_002819) rearrangement: t(4;19)(p16.3;p13.3)(chr4:g.1808676::chr19:g.802509) | Protein Fusion: in frame {FGFR3:PTBP1}	NA
EVT-P-0033176-T01-IM6-142	P-0033176-T01-IM6		RBM20-FGFR3	RBM20	unknown	five_prime	1803168	4	174	False	lost	0.0	False		Protein tyrosine and serine/threonine kinase		FGFR3 (NM_000142) - RBM20 (NM_001134363) rearrangement: t(4;10)(p16.3;q25.2)(chr4:g.1803168::chr10:g.112544434) | Protein Fusion: mid-exon {FGFR3:RBM20}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0036563-T01-IM6-143	P-0036563-T01-IM6		TACC3-FGFR3	TACC3	out-of-frame	five_prime	1804890	7	311	True	lost	0.0	False		Protein tyrosine and serine/threonine kinase		FGFR3 (NM_000142) - TACC3 (NM_006342) Fusion(FGFR3 exon7 fused with TACC3 exon4) : c.931-529:FGFR3_c.305+1845:TACC3dup | Protein Fusion: out of frame {FGFR3:TACC3}	NA
EVT-P-0007131-T01-IM5-144	P-0007131-T01-IM5		TACC3-FGFR3	TACC3	in-frame	five_prime	1808764	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-17 with TACC3 exons 10-16) : c.2275-79:FGFR3_c.1837-97:TACC3dup | Protein fusion: in frame (FGFR3-TACC3)	NA
EVT-P-0028647-T01-IM6-145	P-0028647-T01-IM6		TACC3-FGFR3	TACC3	in-frame	five_prime	1808731	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-17 fused to TACC3 exons 11-16): c.2274+70:FGFR3_c.1942-150:TACC3dup | Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-0030871-T01-IM6-146	P-0030871-T01-IM6		TACC3-FGFR3	TACC3	unknown	five_prime	1808856	17	763	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2288:FGFR3_c.1608:TACC3dup | Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-0000289-T01-IM3-147	P-0000289-T01-IM3		TACC3-FGFR3	TACC3	unknown	five_prime	1806109	8	376	False	lost	0.0	False		Protein tyrosine and serine/threonine kinase		FGFR3 (NM_000142) TACC3 (NM_006342) Duplication (c.1128_c.1592-1871dup) | Protein fusion: mid-exon (FGFR3-TACC3)	NA
EVT-P-0032432-T01-IM6-148	P-0032432-T01-IM6		TACC3-FGFR3	TACC3	in-frame	five_prime	1808674	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2274+13:FGFR3_c.1837-134:TACC3dup | Protein Fusion: in frame {FGFR3:TACC3}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0028321-T01-IM6-149	P-0028321-T01-IM6		TACC3-FGFR3	TACC3	unknown	five_prime	1807884	13	648	False	disrupted	0.638989169675090 2	True			Protein tyrosine and serine/threonine kinase	FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-14 fused to TACC3 exons 7-16): c.1943:FGFR3_c.1591+377:TACC3dup | Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-0027211-T01-IM6-151	P-0027211-T01-IM6		TACC3-FGFR3	TACC3	in-frame	five_prime	1808735	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-17 fused to TACC3 exons 10-16): c.2274+74:FGFR3_c.1836+111:TACC3dup | Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-0027183-T02-IM6-152	P-0027183-T02-IM6		TACC3-FGFR3	TACC3	in-frame	five_prime	1808673	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) Fusion (FGFR1 exons 1-17 fused with TACC3 exons 11-16) : c.2274+12:FGFR3_c.1942-636:TACC3dup | Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-0031155-T01-IM6-153	P-0031155-T01-IM6		TACC3-FGFR3	TACC3	in-frame	five_prime	1808715	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-17 fused with TACC3 exons 8-16. This includes the kinase domain of FGFR3.): c.2274+54:FGFR3_c.1644+125:TACC3dup | Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-0026783-T01-IM6-154	P-0026783-T01-IM6		TACC3-FGFR3	TACC3	unknown	five_prime	1808848	17	760	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-18 with TACC3 exons 11-16) : c.2280:FGFR3_c.1942-78:TACC3dup | Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-0024954-T02-IM6-155	P-0024954-T02-IM6		TACC3-FGFR3	TACC3	in-frame	five_prime	1808755	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-17 fused to TACC3 exons 11-16): c.2275-88:FGFR3_c.1942-76:TACC3dup | Protein Fusion: in frame {FGFR3:TACC3}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0024954-T01-IM6-156	P-0024954-T01-IM6		TACC3-FGFR3	TACC3	in-frame	five_prime	1808755	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-17 fused to TACC3 exons 11-16): c.2275-88:FGFR3_c.1942-76:TACC3dup | Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-0002351-T01-IM3-157	P-0002351-T01-IM3		TACC3-FGFR3	TACC3	unknown	five_prime	1808871	17	768	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) and TACC3 (NM_006342) duplication (71843bp): FGFR3:c.2303_TACC3:c.1620dup | Protein fusion: mid-exon (FGFR3-TACC3)	NA
EVT-P-0024346-T01-IM6-158	P-0024346-T01-IM6		TACC3-FGFR3	TACC3	unknown	five_prime	1808859	17	764	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-18 fused to TACC3 exons 6-16): c.2291:FGFR3_c.1461+69:TACC3dup | Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-0033514-T01-IM6-159	P-0033514-T01-IM6		TACC3-FGFR3	TACC3	unknown	five_prime	1808883	17	772	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-18 fused to TACC3 exons 4-16): c.2315:FGFR3_c.927:TACC3dup | Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-0023261-T01-IM6-160	P-0023261-T01-IM6		TACC3-FGFR3	TACC3	unknown	five_prime	1808927	17	787	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-18 fused to exons 11-16 of TACC3): c.2359:FGFR3_c.1941+673:TACC3dup | Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-0022569-T01-IM6-161	P-0022569-T01-IM6		TACC3-FGFR3	TACC3	in-frame	five_prime	1808705	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-17 fused with TACC3 exons 8-16. This includes the kinase domain of FGFR3.): c.2274+44:FGFR3_c.1645-194:TACC3dup | Protein Fusion: in frame {FGFR3:TACC3}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0020355-T01-IM6-162	P-0020355-T01-IM6		TACC3-FGFR3	TACC3	in-frame	five_prime	1808807	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-18 fused to TACC3 exons 11-16): c.2275-36:FGFR3_c.1942-312:TACC3dup | Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-0019730-T01-IM6-163	P-0019730-T01-IM6		TACC3-FGFR3	TACC3	in-frame	five_prime	1808758	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-17 fused in-frame with TACC3 exons 10-16):c.2275-85:FGFR3_c.1836+25:TACC3dup | Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-0034183-T01-IM6-164	P-0034183-T01-IM6		TACC3-FGFR3	TACC3	unknown	five_prime	1808914	17	782	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-18 fused to TACC3 exons 7-16): c.2346:FGFR3_c.1591+1975dup | Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-0017551-T01-IM5-165	P-0017551-T01-IM5		TACC3-FGFR3	TACC3	unknown	five_prime	1808877	17	770	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-18 fused with TACC3 exons 10-16): c.2309:FGFR3_c.1836+8:TACC3dup | Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-0017444-T01-IM6-166	P-0017444-T01-IM6		TACC3-FGFR3	TACC3	unknown	five_prime	1808864	17	766	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-18 fused with TACC3 exons 11-16): c.2296:FGFR3_c.1941+70:TACC3dup | Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-0017056-T01-IM6-167	P-0017056-T01-IM6		TACC3-FGFR3	TACC3	in-frame	five_prime	1808671	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-17 fused in-frame with TACC3 exons 11-16): c.2274+10:FGFR3_c.1941+881:TACC3dup | Protein Fusion: in frame {FGFR3:TACC3}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0016989-T01-IM6-168	P-0016989-T01-IM6		TACC3-FGFR3	TACC3	in-frame	five_prime	1808718	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-17 fused to TACC3 exons 8-16): c.2274+57:FGFR3_c.1645-139:TACC3dup | Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-0001453-T01-IM3-169	P-0001453-T01-IM3		TACC3-FGFR3	TACC3	unknown	five_prime	1808946	17	793	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase			NA | Protein fusion: mid-exon (FGFR3-TACC3)	NA
EVT-P-0001453-T01-IM3-170	P-0001453-T01-IM3		TACC3-FGFR3	TACC3	unknown	five_prime	1808946	17	793	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase			NA | Protein fusion: mid-exon (FGFR3-TACC3)	NA
EVT-P-0016238-T01-IM6-171	P-0016238-T01-IM6		TACC3-FGFR3	TACC3	in-frame	five_prime	1808706	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exon 17 fused to TACC3 exon 11): c.2274+45:FGFR3_c.1942-654:TACC3inv | Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-0015863-T01-IM6-173	P-0015863-T01-IM6		TACC3-FGFR3	TACC3	unknown	five_prime	1808880	17	771	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-18 fused to TACC3 exons 8-16): c.2312:FGFR3_c.1644+147:TACC3dup | Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-0015195-T01-IM6-174	P-0015195-T01-IM6		TACC3-FGFR3	TACC3	in-frame	five_prime	1808757	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exon17 fused to TACC3 exon8): c.2275-86:FGFR3_c.1645-28:TACC3dup | Protein fusion: in frame (FGFR3-TACC3)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0015024-T01-IM6-175	P-0015024-T01-IM6		TACC3-FGFR3	TACC3	in-frame	five_prime	1808210	15	677	True	disrupted	0.7436823104693141	True			Protein tyrosine and serine/threonine kinase	FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exon 16 fused to TACC3 exon 5) : c.2031-63:FGFR3_c.1385+14:TACC3dup | Protein fusion: in frame (FGFR3-TACC3)	NA
EVT-P-0014530-T01-IM6-176	P-0014530-T01-IM6		TACC3-FGFR3	TACC3	in-frame	five_prime	1808715	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) Fusion (FGFR3 exons 1-17 fused with TACC3 exons 6-16) : c.2274+54:FGFR3_c.1462-52:TACC3dup | Protein fusion: in frame (FGFR3-TACC3)	NA
EVT-P-0014237-T01-IM6-177	P-0014237-T01-IM6		TACC3-FGFR3	TACC3	unknown	five_prime	1808958	17	797	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-18 with TACC3 exons 10-16): c.2390:FGFR3_c.1836+80:TACC3dup | Protein fusion: mid-exon (FGFR3-TACC3)	NA
EVT-P-0012654-T01-IM5-178	P-0012654-T01-IM5		TACC3-FGFR3	TACC3	unknown	five_prime	1808935	17	789	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) -TACC3 (NM_006342)fusion: c.2367:FGFR3_c.1645-121:TACC3dup | Protein fusion: mid-exon (FGFR3-TACC3)	NA
EVT-P-0012373-T01-IM5-179	P-0012373-T01-IM5		TACC3-FGFR3	TACC3	in-frame	five_prime	1808730	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-17 with TACC3 exons 10-16): c.2274+69:FGFR3_c.1837-70:TACC3dup | Protein fusion: in frame (FGFR3-TACC3)	NA
EVT-P-0034666-T01-IM6-180	P-0034666-T01-IM6		TACC3-FGFR3	TACC3	in-frame	five_prime	1808721	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2274+60:FGFR3_c.1941+226:TACC3dup | Protein Fusion: in frame {FGFR3:TACC3}	NA

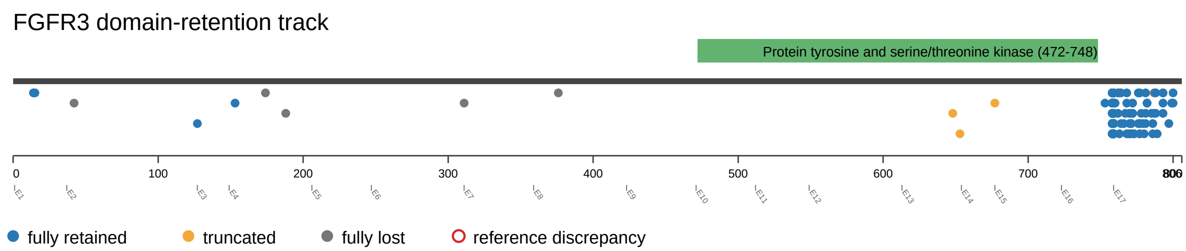
event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0036305-T01-IM6-181	P-0036305-T01-IM6		TACC3-FGFR3	TACC3	unknown	five_prime	1808902	17	778	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 - (NM_000142) TACC3 (NM_006342) Fusion (FGFR3 exon18 fused with TACC3 exon3) : c.2334:FGFR3_c.274:TACC3dup | Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-0008649-T01-IM5-182	P-0008649-T01-IM5		TACC3-FGFR3	TACC3	unknown	five_prime	1808878	17	770	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (exons 1-18 of FGFR3 fused to exons 7-16 of TACC3): c.2310:FGFR3_c.1645-1546:TACC3dup | Protein fusion: mid-exon (FGFR3-TACC3)	NA
EVT-P-0029600-T01-IM6-183	P-0029600-T01-IM6		TACC3-FGFR3	TACC3	in-frame	five_prime	1808697	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2274+36:FGFR3_c.1644+158:TACC3dup | Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-0007082-T03-IM6-184	P-0007082-T03-IM6		TACC3-FGFR3	TACC3	unknown	five_prime	1808872	17	768	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3(NM_000142) - TACC3(NM_006342) Fusion (FGFR3 exon 18 fused to TACC3 exon 8) : c.2304:FGFR3_c.1644+118:TACC3dup | Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-0007082-T02-IM6-185	P-0007082-T02-IM6		TACC3-FGFR3	TACC3	unknown	five_prime	1808872	17	768	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (exons 1-18 of FGFR3 fused to exons 8-16 of TACC3): c.2304:FGFR3_1644+118:TACC3dup | Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-0005966-T01-IM5-186	P-0005966-T01-IM5		TACC3-FGFR3	TACC3	unknown	five_prime	1808853	17	762	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1 to 17 with TACC3 exons 11 to 16) : c.2285:FGFR3_c.1941+423:TACC3dup | Protein fusion: mid-exon (FGFR3-TACC3)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0005862-T01-IM5-187	P-0005862-T01-IM5		TACC3-FGFR3	TACC3	in-frame	five_prime	1808707	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (exons 1-17 of FGFR3 fused to exons 14-16 of TACC3): c.2274+46:FGFR3_c.2223+896:TACC3dup. | Protein fusion: in frame (FGFR3-TACC3)	NA
EVT-P-0004902-T01-IM5-189	P-0004902-T01-IM5		TACC3-FGFR3	TACC3	unknown	five_prime	1808906	17	780	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-18 with TACC3 exons 11-16) : c.2338:FGFR3_:c.1941+300TACC3dup | Protein fusion: mid-exon (FGFR3-TACC3)	NA
EVT-P-0003677-T01-IM5-190	P-0003677-T01-IM5		TACC3-FGFR3	TACC3	unknown	five_prime	1808931	17	788	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase			c.2363:FGFR3_c.1784:TACC3dup | Protein fusion: mid-exon (FGFR3-TACC3)	NA
EVT-P-0003097-T01-IM5-191	P-0003097-T01-IM5		TACC3-FGFR3	TACC3	in-frame	five_prime	1808841	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			NA | Protein fusion: in frame (FGFR3-TACC3)	NA
EVT-P-0036396-T01-IM6-192	P-0036396-T01-IM6		TACC3-FGFR3	TACC3	in-frame	five_prime	1808747	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2274+86:FGFR3_c.162+46:TACC3dup | Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-0016735-T01-IM6-193	P-0016735-T01-IM6		TACC3-FGFR3	TACC3	unknown	five_prime	1808879	17	771	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-18 fused to TACC3 exons 8-16): c.2311:FGFR3_c.1644+71:TACC3dup | Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-0002862-T01-IM3-194	P-0002862-T01-IM3		TACC3-FGFR3	TACC3	unknown	five_prime	1808873	17	769	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase			NA | Protein fusion: mid-exon (FGFR3-TACC3)	NA

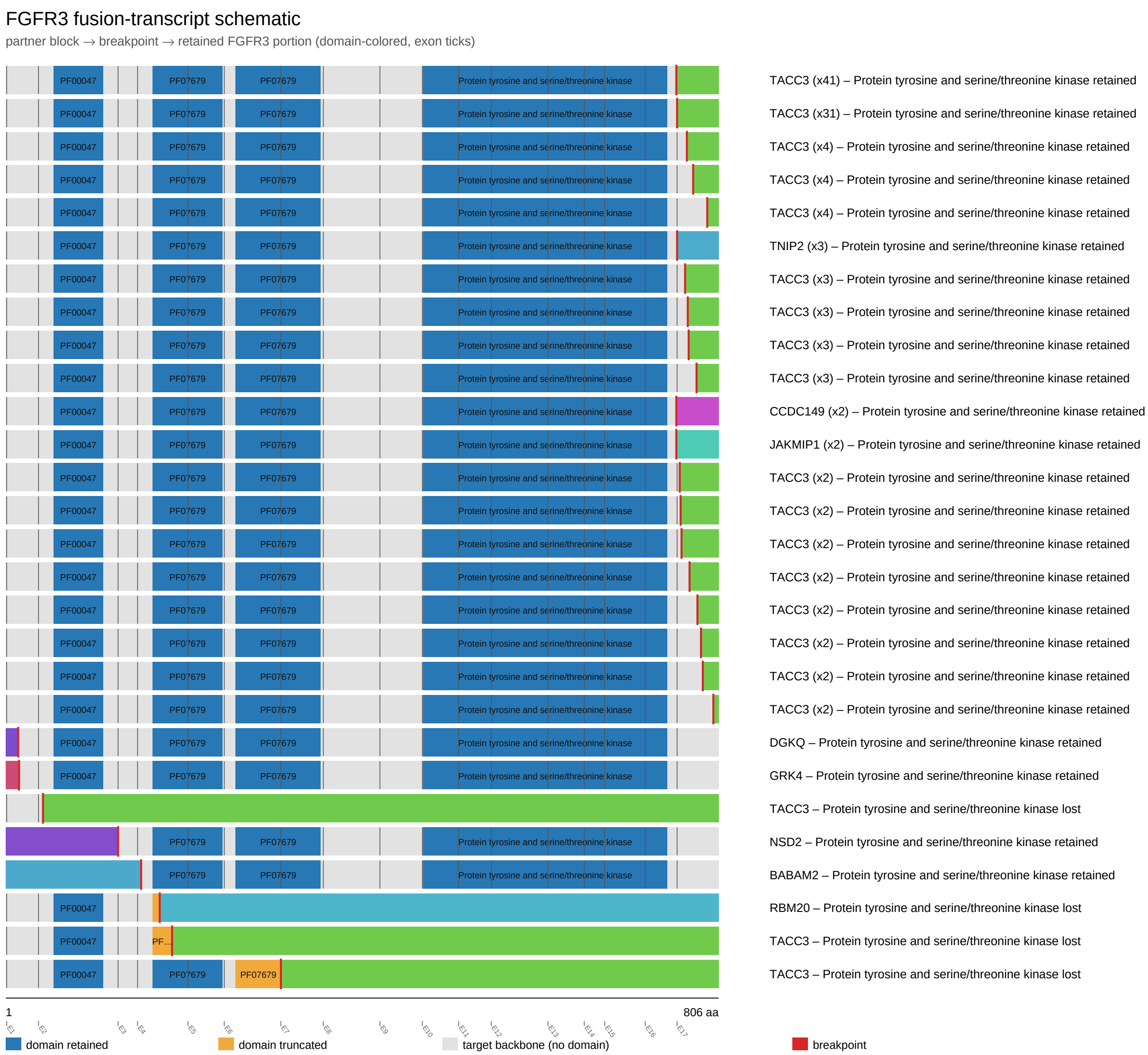
event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0024463-T01-IM6-195	P-0024463-T01-IM6		TACC3-FGFR3	TACC3	in-frame	five_prime	1808801	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-17 fused in-frame with TACC3 exons 10-16):c.2275-42:FGFR3_c.1836+11:TACC3dup | Protein Fusion: in frame {FGFR3:TACC3}	NA

Figures

domain retention outliers

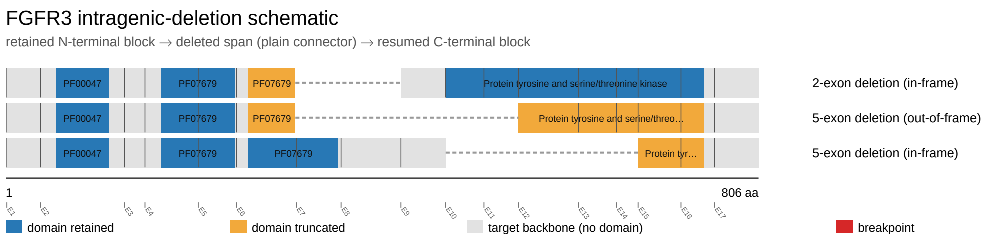


fusion schematic



Showing the top 28 of 52 partner/breakpoint groups by recurrence; see results.tsv for the complete list.

intragenic deletion schematic



reference comparison